SUMMARY · MK 4 OPERATIONS & SUPPLY CHAIN MANAGEMENT · WEEK 4

Project Management

Project management (Ch. 4) with the Whoosh Jakarta–Bandung high-speed rail case and the KOODESIGN example. Covered: why projects are an operations problem, four project types, three organisational structures, the triple constraint with quality in the middle, WBS and milestones, CPM (ES, EF, LS, LF, slack, critical path), three time estimates and completion probability, schedule compression (scope, fast tracking, crashing, overtime, levelling, renegotiation), time–cost models and crashing, earned value management, choosing a delivery method by uncertainty, and five Whoosh discussion questions.

Material

1 Learning objectives and roadmap

1. Explain what a project is and what the triple constraint really costs.
2. Analyse with CPM: critical path, slack, three time estimates.
3. Compress deliberately — fast tracking vs crashing.
4. Evaluate with earned value and match the method to uncertainty.

2 Why project management is an operations problem

A project is **temporary and unique**: a set of related activities with a defined start and end toward a single goal. No learning curve or standard times; every estimate is judgement. Four types: product change, process change, R&D, alliances & partnerships. The **WBS** groups activities into subtasks and work packages.

Uniqueness

Interdependence

Irreversibility

No previous run to copy.

One late activity on the longest chain moves the finish.

Sunk cost is never a reason to continue.

3 How projects are organised: pure, functional, matrix

Structure	How	Strength	Price
Pure	Full-time team, one manager	Clear authority, fast communication	Duplicated resources; specialists drift
Functional	Inside a functional division	Technical depth, careers	Low priority; nobody owns it
Matrix	PM owns what/when; functional managers own who/how	Both strengths	Two bosses

Speed → pure; depth → functional (with explicit priority); both → matrix (settle authority in writing).

4 The triple constraint: scope, time, cost — and quality in the middle

Scope, time and cost are not independent; pulling the date forward implicitly asks for less scope, more money or lower quality.

Dimension	Explanation
Scope	Defined by the WBS; cutting it is the cheapest way to shorten — and usually closed off first.
Time	The longest dependent chain; shortened by fast tracking or crashing, never by exhortation.
Cost	Direct + indirect; the optimum minimises their sum.

! Quality in the middle

When all three are squeezed, quality silently absorbs the difference — name which is fixed, firm and flexible.

5 WBS, milestones and the KOODESIGN example

The WBS is the hierarchy of tasks, subtasks and work packages. Each package needs a duration, predecessor and named resources; milestones are zero-duration decision events; wrong predecessors make a plausible but unexecutable schedule; over-allocation shows at package level.

≡ **KOODESIGN**

19 tasks, 4 milestones; resources named per task with fractional allocations like ID(.5); milestone 3 (asset transfer) is the handover to mass production.

6 The Critical Path Method: ES, EF, LS, LF, slack

CPM finds the **critical path** — the longest chain in time; delay on it delays the project; there can be several.

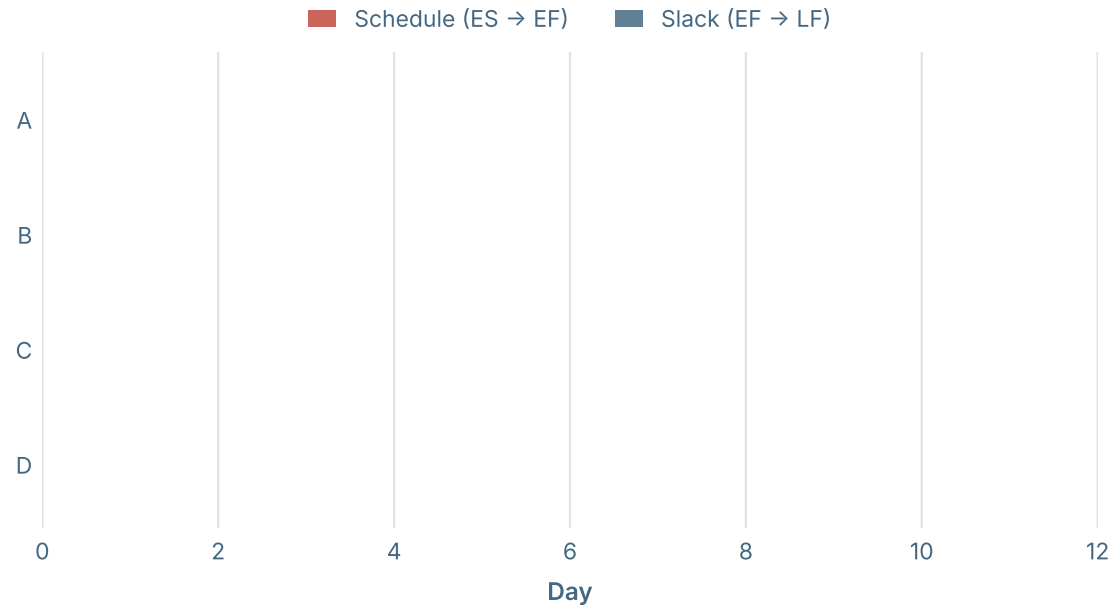
1. Identify activities and durations
2. Sequence predecessors and draw the network
3. Forward pass: ES, EF
4. Backward pass: LF, LS
5. Slack = LS − ES
6. Critical path = zero-slack chain

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Slack = LS − ES = LF − EF
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Activity	Dur	ES	EF	LS	LF	Slack
A	3	0	3	0	3	0

B	4	3	7	5	9	2
C	6	3	9	3	9	0
D	2	9	11	9	11	0

Sample network Gantt (days)



Red = critical (zero slack).

Textbook example

Only E has slack (3 weeks), giving two 38-week critical paths — a fragile project.

7 Three time estimates (PERT) and completion probability

$$TE = (a + 4m + b) \div 6 \cdot \sigma^2 = ((b - a) \div 6)^2$$

Sum TE and variances along the critical path only; the total is roughly normal: $Z = (D - \Sigma TE) \div \sqrt{\Sigma\sigma^2}$.

Slide example

A 33-week critical path gives ~65% chance of finishing within 34 weeks ($Z \approx 0.38$, $\sigma \approx 2.6$ weeks).

Quick practice

a = 2, m = 5, b = 14 → TE = 6, σ^2 = 4.

State the confidence behind a date and show which activities drive the risk.

8 Compressing a schedule: fast tracking, crashing and other options

Technique	How	Cost
Reduce scope	Remove work	Cheapest; usually closed after approval
Fast tracking	Overlap sequential activities	No budget; rework and over-allocation risk
Crashing	Add resources to critical activities	Direct cost up; contract reopened
Supplier overtime	Longer hours, same quality	Small increments (KOODESIGN gained 1 of 5 weeks)
Resource levelling	Delay low-priority tasks	Can lengthen the project
Renegotiate the date	Move the launch	Honest, rarely acceptable
Do nothing	Hope	How projects arrive late

Crash rule

Only on the critical path, cheapest cost per unit of time saved first.

9 Time–cost models and the crashing procedure

Direct costs rise as you compress; indirect costs fall as the project shortens; the optimum minimises the sum.

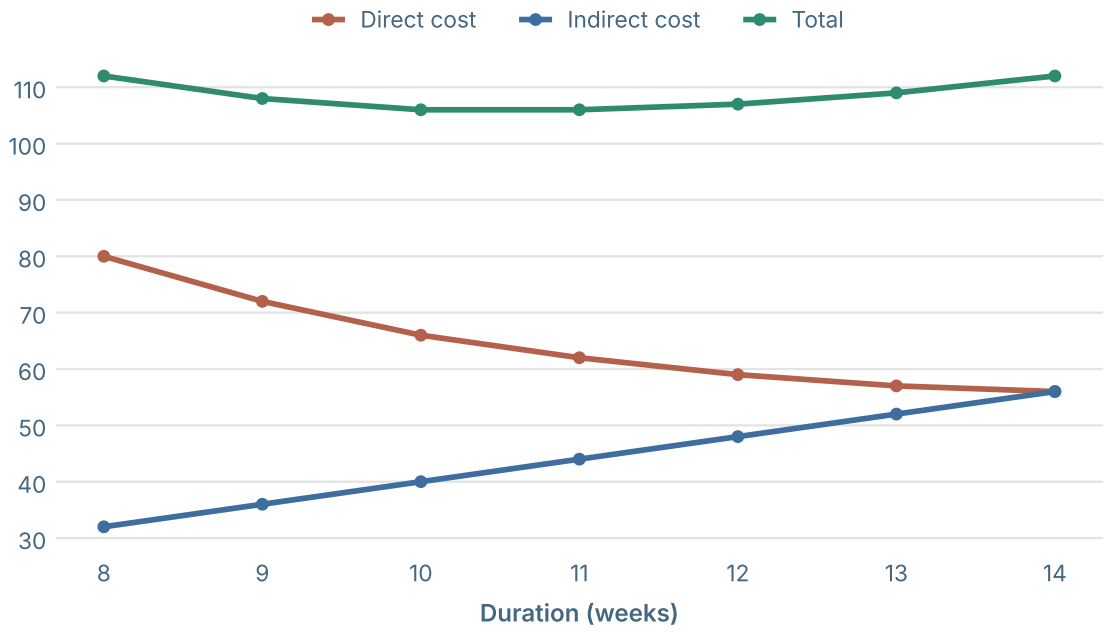
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Cost per period = (crash cost − normal cost) ÷ (normal time − crash time)
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1. CPM with normal and crash times/costs
2. Cost per period for each activity
3. Find the critical path
4. Shorten the cheapest critical activity one period, then recompute

Slope example
6 wk/$10,000 vs 4 wk/$14,000 → $2,000 per week.

In the solved problem, cutting three weeks raised total cost from $94,000 to $97,000.

Time-cost trade-off (illustrative)



10 Earned Value Management (EVM)

EVM combines scope, schedule and cost; spending vs budget means little until you know how much work is done.

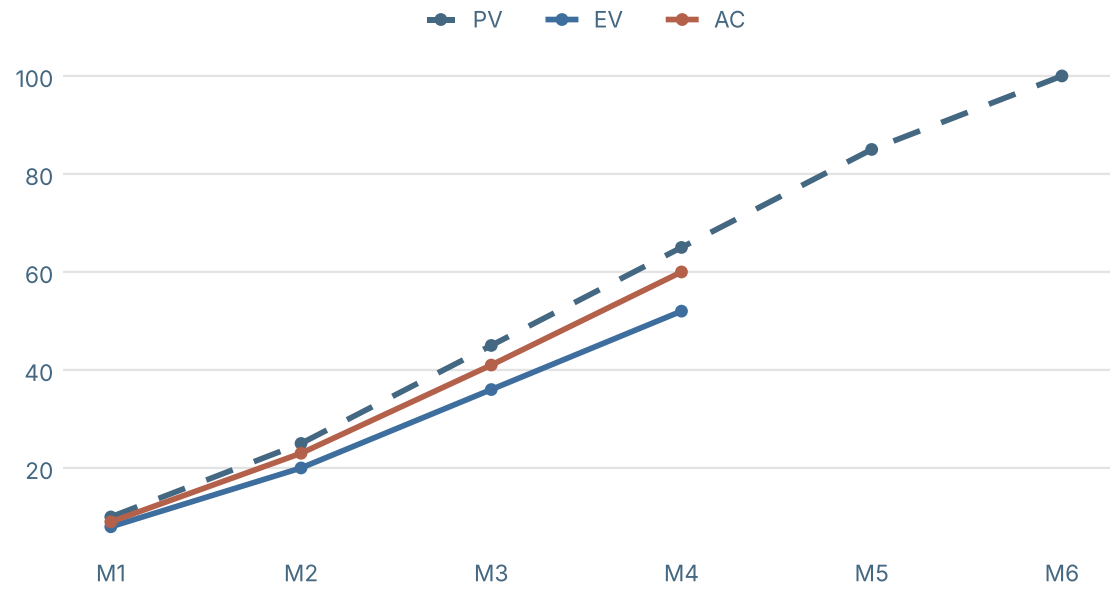
Measure	Alias	Meaning
PV	BCWS	Budgeted cost of work scheduled to date
EV	BCWP	Budgeted cost of work actually performed
AC	ACWP	Actual spend for work performed

$$CV = EV - AC \cdot CPI = EV \div AC \mid SV = EV - PV \cdot SPI = EV \div PV$$

Indexes

Below 1.0: CPI → over budget; SPI → behind schedule.

Cumulative PV, EV, AC (illustrative)



Example: PV 65, EV 52, AC 60 → CPI 0.87, SPI 0.80; EAC ≈ 100 ÷ 0.87 ≈ 115.

11 Choosing the delivery approach by uncertainty

Approach	When it works	Limitation
Waterfall	Stable, regulated requirements	Inflexible; rework piles up late
Agile/Scrum	High change and ambiguity	Hard in hierarchies

Kanban	Constant unpredictable flow	No cadence; needs WIP limits
Lean	Waste elimination	Needs discipline and culture
Hybrid	Fixed market windows	Manager fluent in both
Stage-gate	Research portfolios; kill criteria	Fails if goalposts move
Critical chain	Pooled buffers	Gives up task due dates

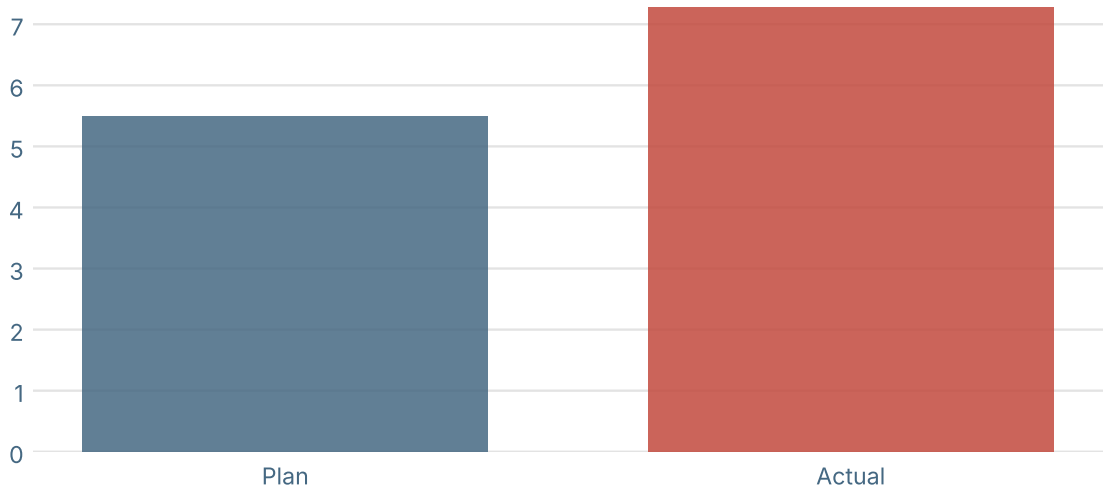
Condition	Method
Simple (clear what, known how)	Waterfall
Complicated (one dimension hard)	Agile for technical, Waterfall or Agile for social/political
Complex (both uncertain)	Scrum
Chaotic	Kanban, stabilise first

Borrow practices safely: keep milestones, limit overlap, measure flow not effort.

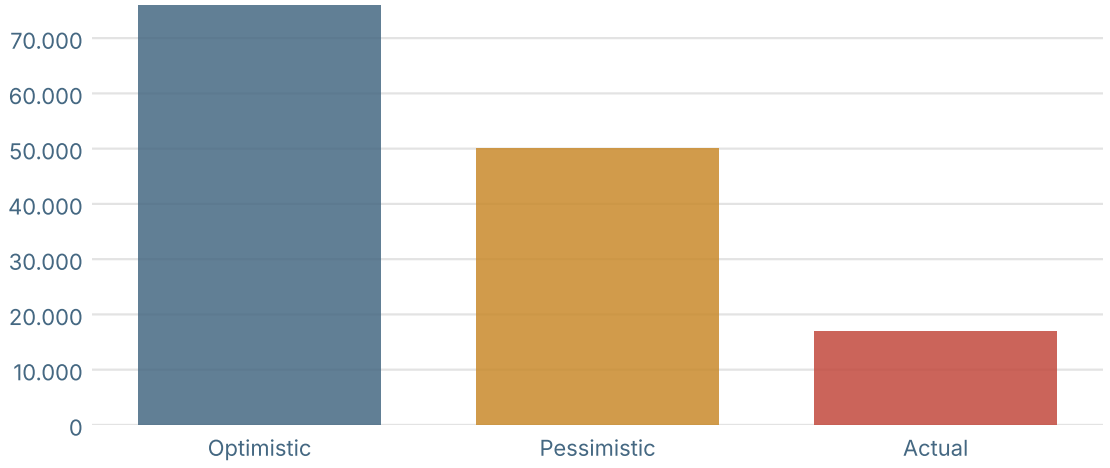
12 The Whoosh case: Jakarta–Bandung high-speed rail

Case C — Whoosh. On 28 August 2026 KCIC's President Director has three weeks before debt management moves to the Ministry of Finance. Cost overrun 31.3% (US$5.5bn → 7.27bn); 4+ years late (2019 target, operations Oct 2023); ~17,000 passengers/day vs a 50,000 pessimistic scenario; 99.9% on-time, 40,000 trips, zero accidents.

Whoosh cost: plan vs actual (US$ bn)



Passengers per day: forecast vs actual



Option	Content	Weakness
A Restructuring	Move debt to the Ministry SMV	Assumptions unrepaired
B Revenue	Chase 30,000 passengers	Growth 2024→25 only 2.4%
C Public asset	Declare social investment	Goalposts move
D Consolidation	Freeze expansion	Read as defeat

Discussion: Q1 scope and cost locked → time slipped and became interest cost on the balance sheet; Q2 much of the US$1.72bn overrun was duration-driven or outside the WBS (US$52m/km vs 17–18m); Q3 a pessimistic case 3× actual is

institutionalised optimism bias; Q4 set kill criteria before investing; Q5 choose, promise measurably, and accept kill criteria for yourself.

13 Wrap-up: six things from the Whoosh case

1. The triple constraint only relocates pressure — on state projects, to the balance sheet.
2. Delay is a financial cost (new debt above 3% vs 2%).
3. Optimistic estimates are a risk (institutionalised optimism bias).
4. Set kill metrics before investing.
5. Project structure is a risk question.
6. Earned value asks a narrow question.

Key takeaways

1. A project is temporary and unique; four types.
2. Pure, functional, matrix structures and their trade-offs.
3. Triple constraint with quality in the middle.
4. WBS, work packages, milestones.
5. CPM: ES, EF, LS, LF, slack, critical path.
6. PERT: TE, variance, Z.
7. Compression options; crash only critical, cheapest first.
8. Crash cost slope; minimum total cost.
9. EVM: PV, EV, AC, CV, SV, CPI, SPI, EAC.
10. Delivery methods by uncertainty.
11. Whoosh lessons.

Case studies

Case & worked answers

Whoosh: the project that could never be stopped

Overrun 31.3% (US$5.5bn → 7.27bn), 4+ years late, ~17,000 passengers/day vs a 50,000 "pessimistic" forecast, yet 99.9% on-time. Q1: with scope, financing and date locked, pressure moved to time and then the balance sheet. Q2: much of the ~US$1.7bn overrun is unplanned scope and duration-driven cost (land prices, interest). Q3: institutionalised optimism bias, no safe channel for bad numbers. Q4: kill metrics (cost +15%, delay >18 months, demand <60% of base) would have triggered around 2019–2021. Q5: e.g. A + D with measurable promises and self-applied kill criteria.

Worked examples

Problem 1 · Medium **PERT dan peluang selesai tepat waktu**

Aktivitas X: optimis 4, paling mungkin 6, pesimis 14 hari. Hitung TE dan variansnya. Jika jalur kritis total TE = 20 hari dengan total varians 4, berapa peluang proyek selesai ≤ 23 hari?

1. `TE = (a + 4m + b) ÷ 6 = (4 + 24 + 14) ÷ 6 = 7 hari`
2. $\sigma^2 = ((b - a) \div 6)^2 = (10 \div 6)^2 = 2{,}78$
3. Jalur kritis: σ = √4 = 2 hari.
4. z = (23 − 20) ÷ 2 = 1,5 → dari tabel normal, P(z ≤ 1,5) = 0,933.

Answer: TE 7 hari, varians 2,78; peluang selesai ≤ 23 hari ± 93%.

Problem 2 Easy **Biaya crashing**

Aktivitas normal 10 hari (Rp20 juta), bisa dipercepat jadi 7 hari (Rp29 juta). Berapa biaya percepatan per hari?

1. `Biaya/hari = (crash cost − normal cost) ÷ (normal time − crash time)`
2. = (29 − 20) ÷ (10 − 7) = 9 ÷ 3 = 3.

Answer: Rp3 juta per hari. Percepat hanya jika aktivitas ini kritis dan termurah di antara aktivitas kritis.

Problem 3 Medium **Earned value**

BAC Rp500 juta. Saat ini PV = 200, EV = 180, AC = 210. Hitung CV, SV, CPI, SPI, dan EAC.

1. CV = EV − AC = 180 − 210 = −30 (boros).
2. SV = EV − PV = 180 − 200 = −20 (terlambat).
3. CPI = 180 ÷ 210 = 0,857; SPI = 180 ÷ 200 = 0,90.
4. `EAC = BAC ÷ CPI = 500 ÷ 0,857 = 583`

Answer: Over budget dan terlambat; perkiraan biaya akhir ± Rp583 juta.

Practice questions

Try answering first. The answer key is below each question.

1. Zero slack means...

A. Freely delayable
B. On the critical path
C. Finished
D. Unimportant

Answer: B. Critical path.

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2. a 2, m 4, b 12. TE = ...

A. 4
B. 5
C. 6
D. 6.5

Answer: B. 30/6.

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3. When crashing, first compress...

A. Cheapest anywhere
B. Cheapest critical activity
C. Most slack
D. All at once

Answer: B. Non-critical crashing buys nothing.

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4. EV 52, AC 60, PV 65. Status...

A. Under budget, ahead
B. Over budget, behind
C. Under budget, behind
D. Over budget, ahead

Answer: B. Both indexes below 1.

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5. With scope, financing and date locked, pressure finally landed on...

A. Technical quality
B. Time, then the balance sheet
C. Stations
D. Nothing

Answer: B. 4 years late, cost overrun.

Sources

- P3MD MK4 OSCM — Project Management (class slides, final)
- Jacobs & Chase — OSCM Ch. 4; KOODESIGN (HKUST ST131); Whoosh (SBM ITB Case Series)